

lexical head: learned weights of hashed sol_k unigrams + bigrams $\rightarrow \ell_k$

$$\rightarrow Z_k = S_k + \ell_k$$

goal $\oplus$ sol₁

**embedding
+ difference tag**

Transformer
encoder (shared)

**cross-solution
attention
(ESIM fusion)**

**diff-guided
pooling**

scorer $\rightarrow s_1$

difflib alignment: shared / differing tokens

softmax
(z_1, z_2)

goal $\oplus$ sol₂

**embedding
+ difference tag**

Transformer
encoder (shared)

**diff-guided
pooling**

scorer $\rightarrow s_2$

orange: our PIQA-specific components (ablated in Table 2); blue: standard components